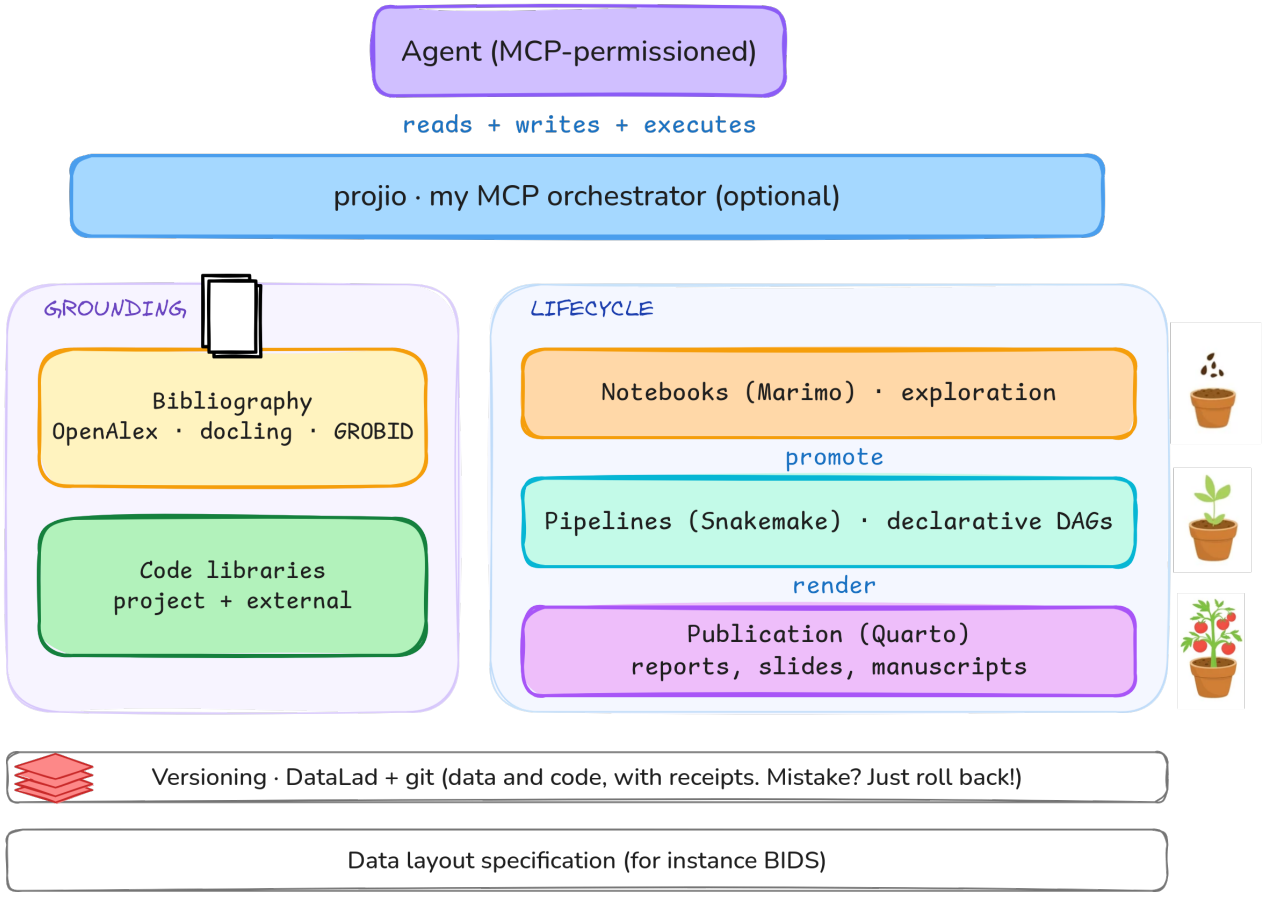
Open-science conventions as agent infrastructure

A stack-aware workflow experiment. ***Proposed contribution to the LSM Future Skills 2026 breakout,*** **\ AI in Daily Research: Sharing What Works (and What Doesn't).** ***22 September 2026***

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Agentic workflows operate in a project workspace, reading and writing across the same files, pipelines, notes, and references a researcher uses. Their effectiveness on any given project depends heavily on the structure of that workspace.

When a project follows mature conventions, an agent navigates it with little prompting and writes its output back into the same structured form. The conventions do most of the work.



Each of these tools is mature in its own right. Together they form a ***project surface*** *an agent can read and write across without leaving the repository.*

What the agent works with

Four structured pieces a project exposes when organized this way.

- **Memory.** A structured set of project notes (ideas, tasks, results, meeting records, decisions on disk) sitting next to an activity log of every tool call. The reasoning behind any later claim is reconstructible by reading the project's git history.
- **Grounding in literature and existing code.** A curated project bibliography and a catalog of code libraries (project plus external mirrors) that the agent consults while writing. Claims get linked to citations as they appear; new code builds on existing libraries.
- **Exploratory power and rerunnable pipelines.** Interactive notebooks, for example marimo with HoloViews/Bokeh. The agent can author and edit these notebooks alongside the researcher in real time. Once the researcher decides to apply the method across the database, the agent can be prompted to promote the program into a pipeline (Snakemake with config files).
- **Communication.** Quarto reports, revealjs slide decks, and eventually manuscript sections all re-render from the same data, pipelines, and bibliography. The same project tree, including reports, can be published as a GitHub or GitLab Pages site that an agent keeps up to date for collaborators.

Still brittle

- Agents produce structure faster than researchers can maintain it.
- Context retrieval still needs human checking.

Open-science structure makes a project legible to agents. The researcher stays in charge of the research.

References

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